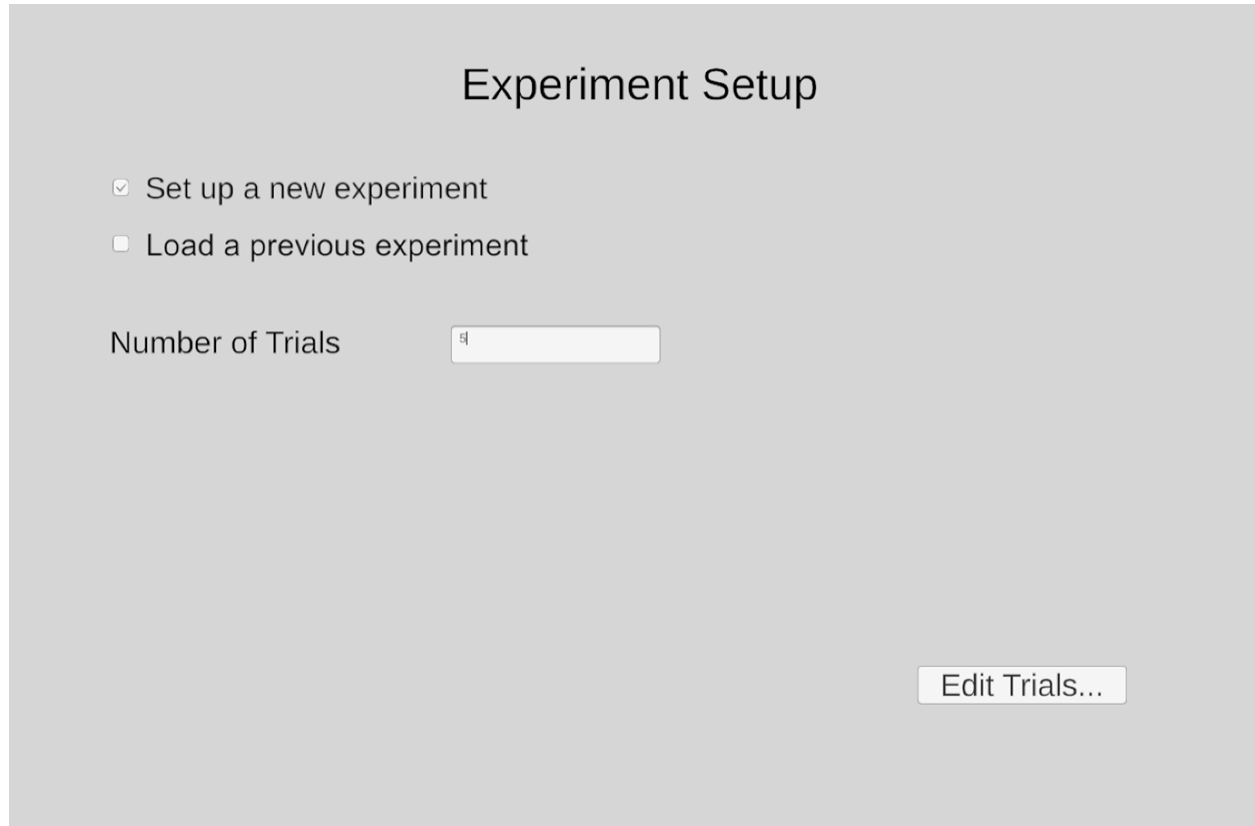


To set up an experiment (configure the parameters):

1. Click on “Set up a new experiment”
2. Input the number of trials that this experiment should have.
3. Click “Edit Trials...”



The screenshot shows a web interface titled "Experiment Setup" in a large, bold, black font. Below the title, there are two radio button options: "Set up a new experiment" (which is selected with a checkmark) and "Load a previous experiment". Below these options, the text "Number of Trials" is followed by a text input field containing the number "5". In the bottom right corner of the interface, there is a button labeled "Edit Trials..."

You will then see a page like this:

Trial 1 of 1

Number of quadrants (1-10)

Radius (m)
(Recommended: 20)

Number of Proximal Cues (0-8)

Trial Type

Time Limit (s)
(-999 for no time limit)

Distal Cues ☒

Treasure Location

Cue Location

Starting Location

Cue Selection

1 ☐

2 ☐

3 ☐

4 ☐

5 ☐

6 ☐

7 ☐

8 ☐

Cue offset: 2.0 m

File name (will default to exp_x.json)

(Press "0" to quit)

Explanations for selected parameters:

Number of quadrants	1-10	Number of quadrants tracked in the data files
Trial Type	Visible	Treasure chest location is visible.
	Invisible	Treasure chest location is hidden.
	Absent	No treasure chest exists.
Treasure Location	Randomize	Treasure location is randomized.
	Manual	Pick the treasure chest location by clicking within the circle, indicated by a gold dot.
Cue Location	Auto	Proximal cues are spaced equally apart.
	Manual	Pick the proximal cues by clicking the circle, indicated by blue dots.
Cue Offset	2.0m - 50.0m	Distance of proximal cues from the perimeter, from 2.0m to 50.0m.
Starting Location	Auto	Ensures that the starting location of the player is not in the same quadrant as treasure chest, and the player does not face the treasure chest.
	Randomize	Randomizes the starting location of player with no restrictions.
	Manual	Pick the starting location of the player, indicated by a green dot.

A few more buttons:

1. Copy Settings: This copies settings from the current trial to all other trials.

Cue selection:

You can specify exactly which proximal cues are displayed by selecting the box corresponding to each cue's number. In order from 1 to 8:

1. park_statue
2. park_ice_cream_truck
3. park_bench
4. park_sign
5. park_fountain
6. park_Gazebo
7. park_drinking_fountain
8. park_trashcan

Finally, you must save the settings to a file to run it in the future. Give the file a name, then press "save settings".

Data Files

Data is saved in the "ExpData" folder located in the root of the project.

PathTrace_{id}_trial{number}.jpeg	Traces the path taken by the player. The green circle indicates the starting location, and the X indicates the treasure chest location. The number of quadrants drawn here corresponds to the setting selected in trial settings. For our study, the number of quadrants should always be 4.
PositionData_{id}.csv Note, bolded words refer to the actual title of the columns for these variables in the csv file.	Contains: • the number of the trial, trialNumber • every .10 sec, the participant's position is sampled across each trial, the time at which this position is sampled is recorded as time • Coordinates of participant across all trials. PositionX, PositionZ. • Heading in degrees of the participant, HeadingDeg

	Distance of participant's starting point from treasure chest in meters per trial, DistanceToChest
TrialData_{id}.csv Note, bolded words refer to the actual title of the columns for these variables in the csv file.	Contains: - the number of the trial, trialNumber - Total distance covered by player in meters per trial, totalDistance - Total time per trial taken to find chest in seconds, totalTimeTaken - participant's average distance to the treasure over the course of the trial (average proximity) calculated continuously from start until they find it or time runs out, averageDistanceToChest - Number of quadrants selected. For our study, this should be 4. numberOfQuadrants - Designates which quadrant contains the treasure chest. chestQuadrant - The amount of time (in sec) spent in quadrant 1, timeQ1 - The amount of time (in sec) spent in quadrant 2, timeQ2 - The amount of time (in sec) spent in quadrant 3, timeQ3 - The amount of time (in sec) spent in quadrant 4, timeQ4 - Proportion of time spent in the quadrant containing the treasure chest, proportionInChestQuadrant fractalDimensionBox. This is a measure of path complexity. A higher value indicates a more random, circuitous path. A lower value indicates a more direct, efficient path. Fractal dimensionality is bounded by two values 1 (equating to a smooth line) and 2, implying fractal complexity of Brownian motion (Singh et al., 2016). Fractal dimensionality of path is calculated by the box counting method from Foroutan-pour et al. (1999). 1 means the trajectory behaved like a simple line/curve at the measured scales. That could be a straight-ish path, a smooth arc, or a path with too

	little spatial complexity to increase box coverage beyond line-like behavior. ○ > 1 means the path occupied 2D space more densely than a simple line. Loops, repeated searching, zig-zags, spirals, backtracking, or broad wandering can all push it above 1. ● Straightness Index – the total displacement (i.e. distance from starting location to treasure chest) divided by total distance covered during trial (Codling et al., 2008), straightnessIndex ● Tortuosity in radians per meter. A measure of how direct the path taken to the treasure chest is. tortuosityRadPerMeter Traditionally, it is calculated as the ratio of the actual distance the participant traveled to the treasure chest (path length) to the shortest distance possible between the starting position and the treasure chest location (Singh et al., 2016). ● Angular entropy –a measure of disorder or unpredictability in a participant’s path. a lower value indicates less randomness in turns; a higher value indicates higher randomness in turns. This is calculated by first computing Shannon entropy across the 18 movement-angle bins, then dividing by the maximum possible entropy for 18 bins: $\log_2(18) = 4.17$ bits. This normalizes the value to a 0-1 scale, where 0 means all movement is in one angular bin and 1 means movement directions are evenly distributed across bins (Shannon, 1948; Pielou, 1966). angularEntropyBits ○ For instance, we would expect a lower angular entropy score for participants who stays closer to the ideal route to the treasure chest, Angular entropy is a metric of how much the heading direction changes for a given participant path.
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	• Crossings – (Probe trial only): number of crossings over the treasure chest in path taken during the trial, crossings
QuadrantData_[id].csv Note, bolded words refer to the actual title of the columns for these variables in the csv file.	Contains: • the number of the trial, trialNumber • the number of quadrants in the environment. For the SNAP study, this should always be 4. numberOfQuadrants • This is the quadrant where the treasure chest is actually located in each trial, chestQuadrant • The amount of time (in sec) spent in quadrant 1, timeQ1 • The amount of time (in sec) spent in quadrant 2, timeQ2 • The amount of time (in sec) spent in quadrant 3, timeQ3 • The amount of time (in sec) spent in quadrant 4, timeQ4 • The proportion of time spent in the quadrant that contains the treasure chest, proportionInChestQuadrant

To run an experiment:

From the first page, click “Load a previous experiment”, select the settings file from the dropdown menu, input the participant ID, then press “Run...”

Experiment Setup

☐ Set up a new experiment

☒ Load a previous experiment

Participant ID

Settings Choice

Run...

To quit the trial at any time, press “0”.

To exit the set-up screen, hit the “x” in the right-hand corner.

Github:

<https://github.com/wristy/vMorrisWaterTask>

Works Cited

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